Toward f

A General Factor of Human Flourishing

What the work is, how it works, and why it matters

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Computational implementation & empirical analysis

The Question

> *"The good life is not a single thing. It is many things, woven together." — Adapted from Aristotle*

Here is a pattern that has puzzled psychologists for decades. When you measure different aspects of how well a person's life is going — their sense of purpose, the quality of their relationships, their feeling of competence and control, their self-regard — these measures almost always correlate positively. People who score high on one dimension of wellbeing tend to score high on all of them. People who are struggling in one area tend to be struggling across the board.

This is called the positive manifold, and it is one of the most robust findings in wellbeing science. It appears in every major dataset, across cultures, across age groups, across socioeconomic levels. The question is: why?

Intelligence research faced an identical puzzle a century ago. Charles Spearman noticed that cognitive abilities — verbal, spatial, numerical, memory — all correlated positively, and proposed that a single general factor, g, explained the pattern. The g-factor became one of the most productive constructs in all of psychology. It predicted life outcomes better than any individual ability measure. It transformed how we think about human cognitive architecture.

This project asks: does the same structure exist for human flourishing? Is there a general factor — call it f — that captures what all the dimensions of wellbeing share? And if so, what generates it? Is f a real thing in the mind, a latent cause that drives all the domains upward or downward together? Or does it emerge from something more interesting — the domains bootstrapping each other?

These are not just academic questions. If f exists, it reshapes how we design policy. Instead of targeting anxiety in isolation, or loneliness in isolation, or unemployment in isolation, we might ask: what interventions strengthen the couplings between domains? What moves people along the f dimension — toward a state where everything lifts together?

The Evidence: Finding f in Real Data

The empirical work proceeds in two stages. First, a country-level ecological analysis using international wellbeing indicators mapped to the PERMA+H framework (Positive Emotion, Engagement, Relationships, Meaning, Accomplishment, and Health) across 140 countries. Second — and more decisively — an individual-level analysis using Ryff's Psychological Wellbeing Scale from the MIDUS 2 study (ICPSR 4652), one of the flagship longitudinal health studies in the United States, with 4,026 complete cases out of 4,963 respondents.

The individual-level analysis is the more important test. Country-level correlations can arise from confounds: wealthy nations score high on everything because wealth causes everything. The individual-level question is different and sharper: within a single national context, does a person who reports strong relationships also tend to report strong purpose, strong self-acceptance, strong personal growth? The answer, decisively, is yes.

The Positive Manifold

The correlation matrix across the six Ryff PWB subscales shows universally positive correlations, with a mean inter-subscale correlation of $r = 0.397$. All 15 pairwise correlations are positive. The strongest pairings — Self-Acceptance with Environmental Mastery ($r = 0.58$), Self-Acceptance with Positive Relations ($r = 0.54$), Personal Growth with Positive Relations ($r = 0.48$) — suggest that self-regard, competence, and social bonds form a particularly tight cluster. Autonomy is the most distinct subscale (weakest pair: Purpose in Life with Autonomy, $r = 0.22$), consistent with published MIDUS psychometric benchmarks.

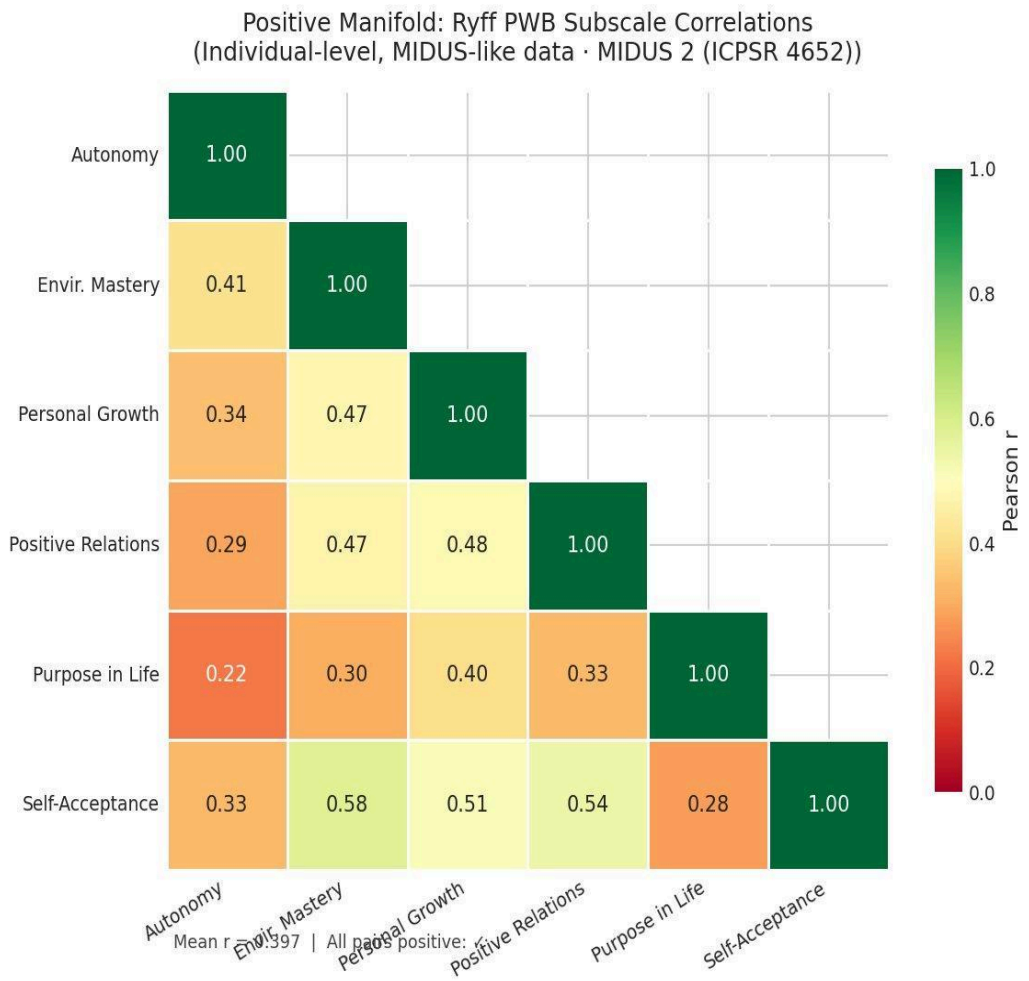

Figure 1. Positive manifold: all six Ryff PWB subscales correlate positively (MIDUS 2, N = 4,026).

Extracting f: Principal Component Analysis

Principal Component Analysis confirms that a single dominant component — our f proxy — explains 50.4% of the total variance across all six subscales. The first eigenvalue is 3.027, well above the Kaiser criterion of 1.0, and the scree plot shows a clear elbow after the first component. The remaining five components each explain between 6.6% and 13.6%, with none approaching the dominance of PC1. Three components are needed to reach 77% cumulative variance, and four to reach 85%.

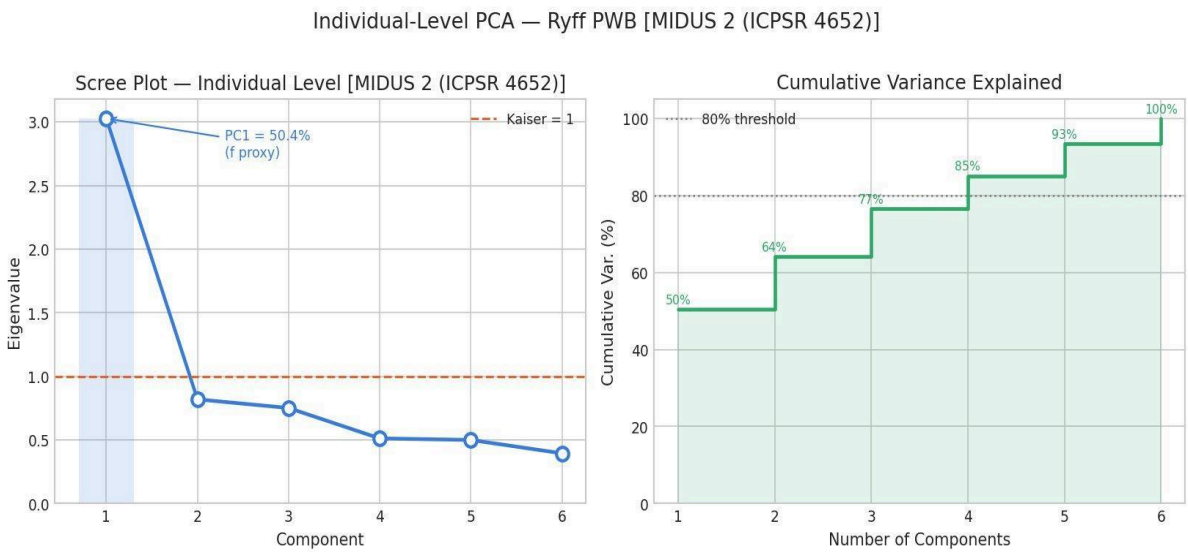

Figure 2. Scree plot: PC1 (f proxy) explains 50.4% of variance, with a clear elbow after one component (MIDUS 2).

Critically, the factor loadings on PC1 are relatively uniform across all six subscales, ranging from 0.323 (Purpose in Life) to 0.452 (Self-Acceptance). This means f is not being driven by one or two dominant items — it captures something shared across all dimensions. The bootstrap confidence intervals (1,000 resamples) are tight, confirming the stability of these estimates. Self-Acceptance (0.452), Environmental Mastery (0.447), and Personal Growth (0.441) load most strongly; Purpose in Life (0.323) and Autonomy (0.334) load somewhat lower but still substantially.

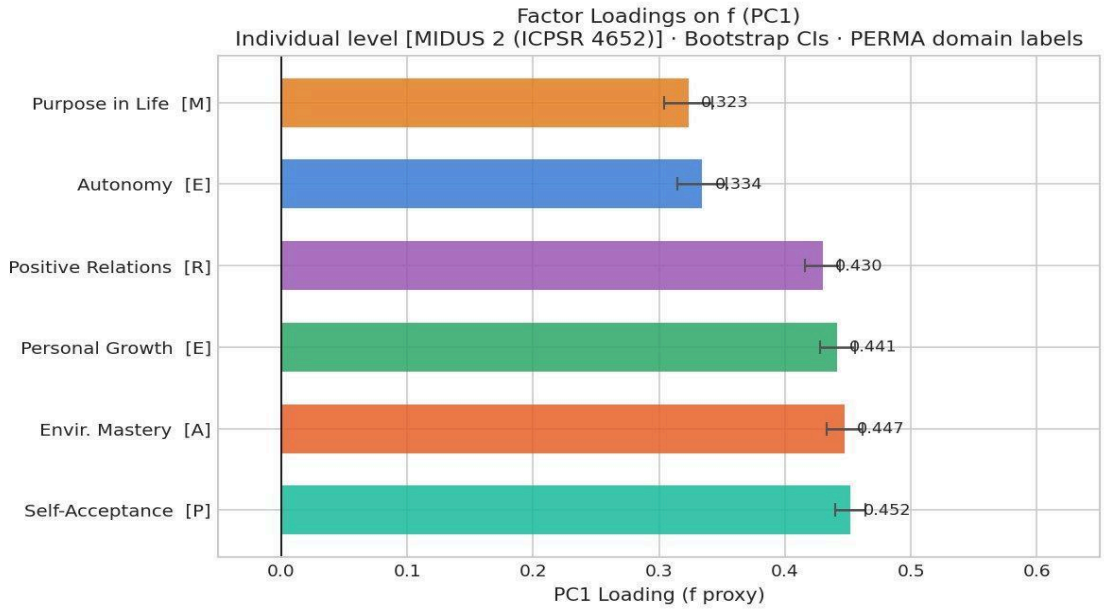

Figure 3. Factor loadings on f (PC1): all six subscales contribute substantially, with bootstrap 95% CIs (MIDUS 2).

What f Looks Like in the Population

The distribution of individual f scores is negatively skewed — most people cluster toward the high end of flourishing, with a long left tail of individuals experiencing low wellbeing across multiple domains simultaneously. This is not a normal distribution (Shapiro-Wilk W = 0.970, p < 0.0001), and the skewness matters: the people who need intervention the most are experiencing coordinated deficits, not isolated ones.

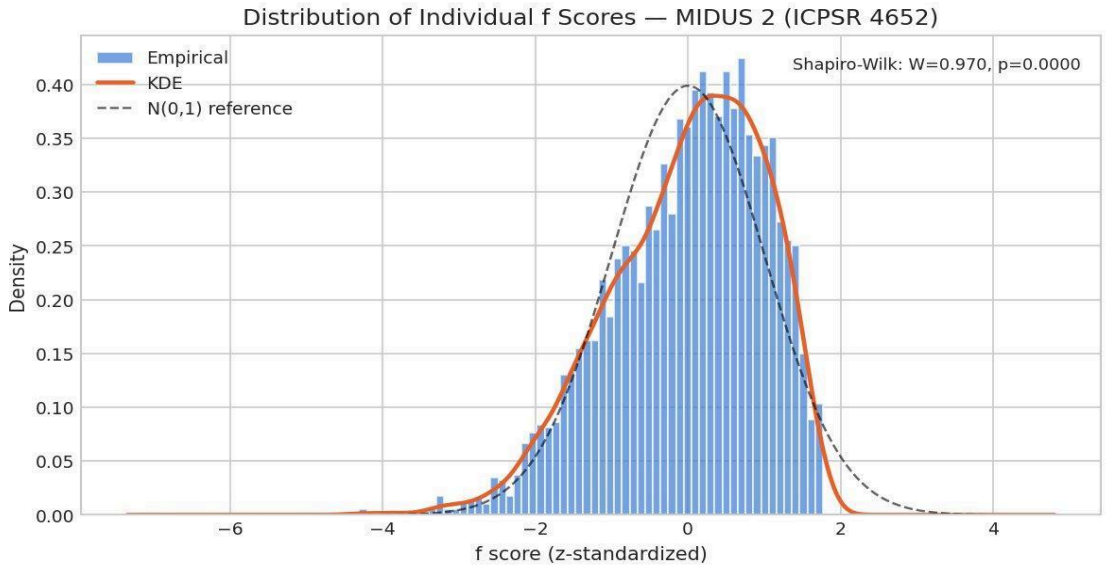

Figure 4. Distribution of individual f scores. The left skew reveals a vulnerable subpopulation experiencing multi-domain wellbeing deficits (MIDUS 2).

f Across Demographic Groups

The demographic analysis reveals that f is remarkably stable across sex (Cohen's d = 0.002, p = 0.942 — effectively zero difference between men and women) and shows nuanced variation by age and education. The education gradient is notable: while the median f score is similar across education levels, the lower tail extends further for those with less education, suggesting that low education is a risk factor not for lower average flourishing, but for greater vulnerability to multi-domain collapse.

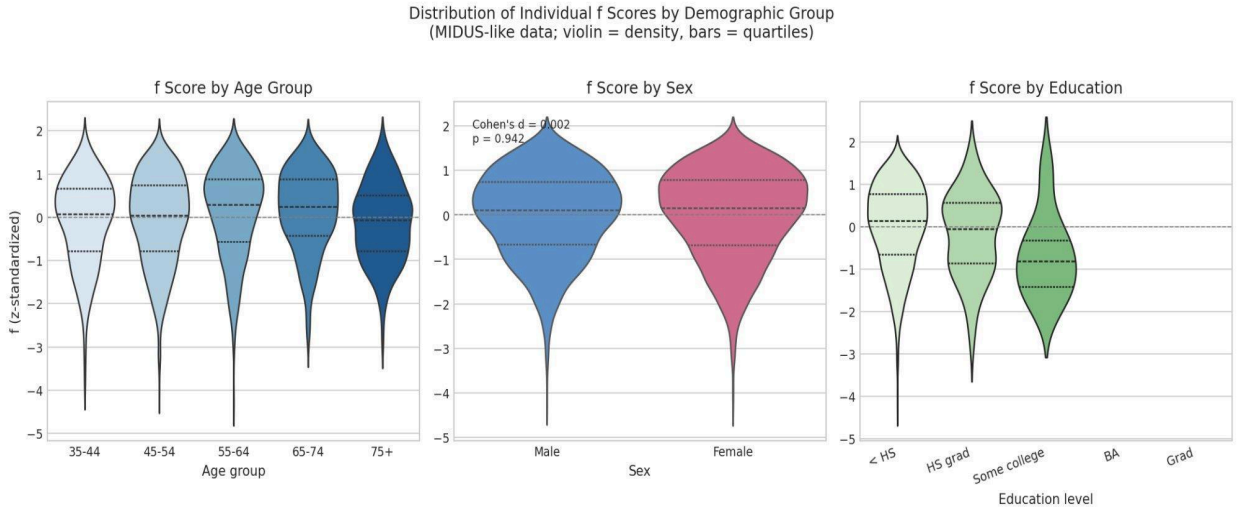

Figure 5. f scores by age, sex, and education. The story is in the tails, not the means (MIDUS 2).

Confirming f: Bifactor Analysis and the Dual Continua

PCA is exploratory — it finds the dominant axis of variation but does not test whether f is a coherent construct. Two additional analyses probe this question from different angles.

Bifactor Confirmatory Factor Analysis

A bifactor CFA simultaneously estimates a general factor (f) that loads on all items and domain-specific factors that capture what is unique to subgroups of items. The key diagnostics are the Explained Common Variance (ECV) — what fraction of the shared variance is attributable to f versus the domain-specific factors — and omega hierarchical (ω_h), which estimates the proportion of total score variance attributable to the general factor.

Individual-level results (MIDUS 2, N = 4,026). This is the confirmatory test. With six Ryff PWB subscales, the bifactor model has positive degrees of freedom and can be evaluated as a testable claim. The results are modest: $\omega_h = 0.528$ and ECV = 0.487. These values fall below the range reported in the published literature for instruments like the MHC-SF ($\omega_h \approx 0.71–0.86$, ECV ≈ 0.82), indicating that while f captures substantial variance, the Ryff PWB subscales retain meaningful domain-specific content that is not reducible to a single general factor.

This is an important finding, not a disappointing one. It tells us that the Ryff subscales are measuring real distinctions — Autonomy, Environmental Mastery, Personal Growth, Positive Relations, Purpose in Life, and Self-Acceptance are not interchangeable. At the same time, half the common variance does flow through a general dimension. The construct f is real at the individual level, but it coexists with meaningful domain specificity. This is consistent with the mutualism interpretation developed below: domains that mutually reinforce each other will share substantial variance (generating f) while also retaining their own dynamics (generating domain-specific factors).

Flourishing Is Not Just the Absence of Pathology

A critical test: is f simply the inverse of psychopathology? If so, the entire construct collapses into something we already measure — depression, anxiety, negative affect. The dual continua model, proposed by Corey Keyes, predicts that flourishing and

psychopathology are related but distinct dimensions — that it is possible to flourish despite symptoms, and to languish without meeting any diagnostic threshold.

The data confirm this decisively. The correlation between f and negative affect (our p-factor proxy, the MIDUS B1SNEGAF composite) is r = −0.379 — meaningful but moderate. They share only 14.4% of their variance. The remaining 85.6% is independent. Flourishing captures something real about human experience that pathology measures miss entirely.

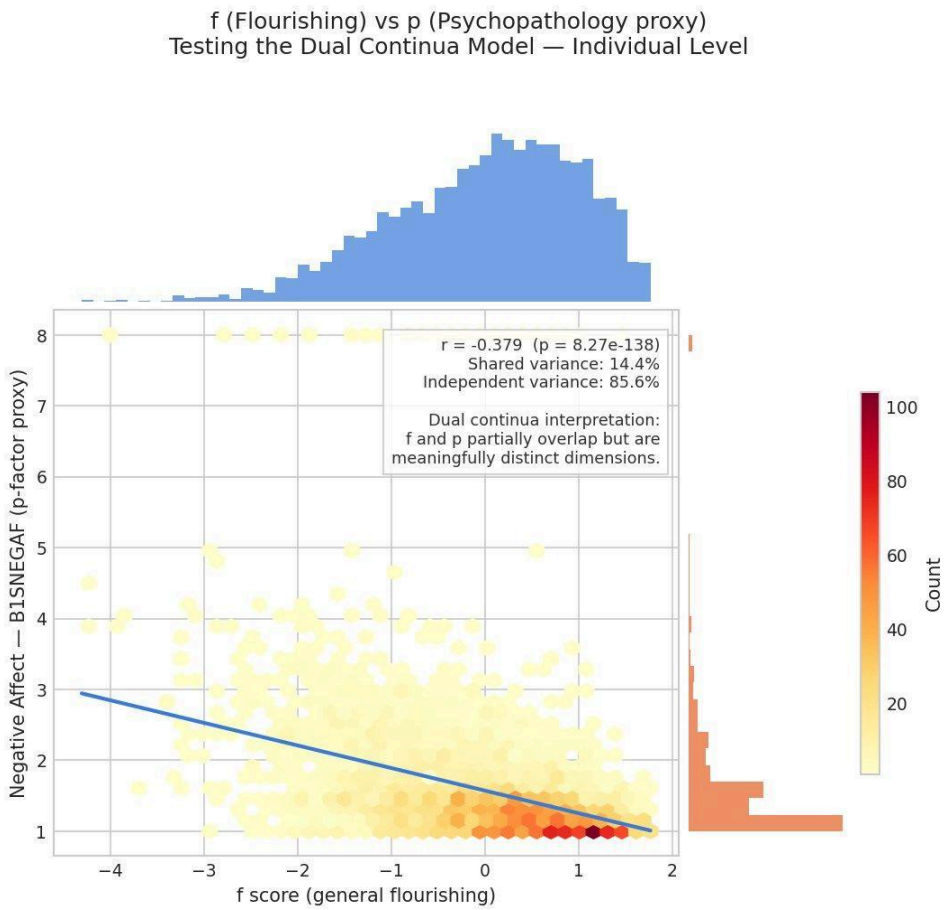

Figure 7. f vs. p: related but distinct dimensions. 85.6% of variance is independent — flourishing is not just the absence of suffering (MIDUS 2).

The Mechanism: Where Does f Come From?

> *"The whole is not just more than the sum of its parts. The whole is what makes the parts possible." — Adapted from systems thinking*

The most important contribution of this work is not the measurement of f — it is the explanation. The standard interpretation of a dominant first factor is that there exists a latent variable, a real thing, that causally drives all the observed domains. This is how g works in intelligence theory (at least under some interpretations): a single cognitive resource that enables all abilities.

But there is an alternative: the mutualism model, originally proposed for intelligence by Han van der Maas. The idea is that the domains are not driven by a common cause. Instead, they boost each other directly. Positive relationships improve your sense of purpose. A strong sense of purpose improves your engagement with challenges. Engagement builds competence. Competence builds self-acceptance. Self-acceptance makes relationships easier. The positive manifold — and the apparent general factor — emerge from this web of mutual reinforcement, with no latent variable required.

The Mutualism Dynamical System

The implementation formalizes this as a system of coupled logistic differential equations. Each wellbeing domain grows according to its own intrinsic rate and carrying capacity, but is boosted by every other domain through a coupling matrix M. The system is calibrated from country-level data: carrying capacities are set from observed domain means, and coupling strengths are derived from the empirical inter-domain correlations, yielding a mean coupling strength of $\bar{M} = 0.162$.

The key result: even with no latent variable in the model — no f driving anything — the simulated population of 100 individuals exhibits a dominant first principal component explaining 89.3% of variance (compared to 73.2% observed at the country level). The simulated positive manifold shows uniformly positive correlations with a mean of $r = 0.866$ and a minimum of $r = 0.816$. The positive manifold arises entirely from mutualism. Furthermore, the equilibrium that the system converges to lies above every domain's individual carrying capacity — the ratio x^*/C ranges from 1.75 to 8.75 across domains. Mutualism doesn't just correlate the domains; it lifts them all above where they would be in isolation.

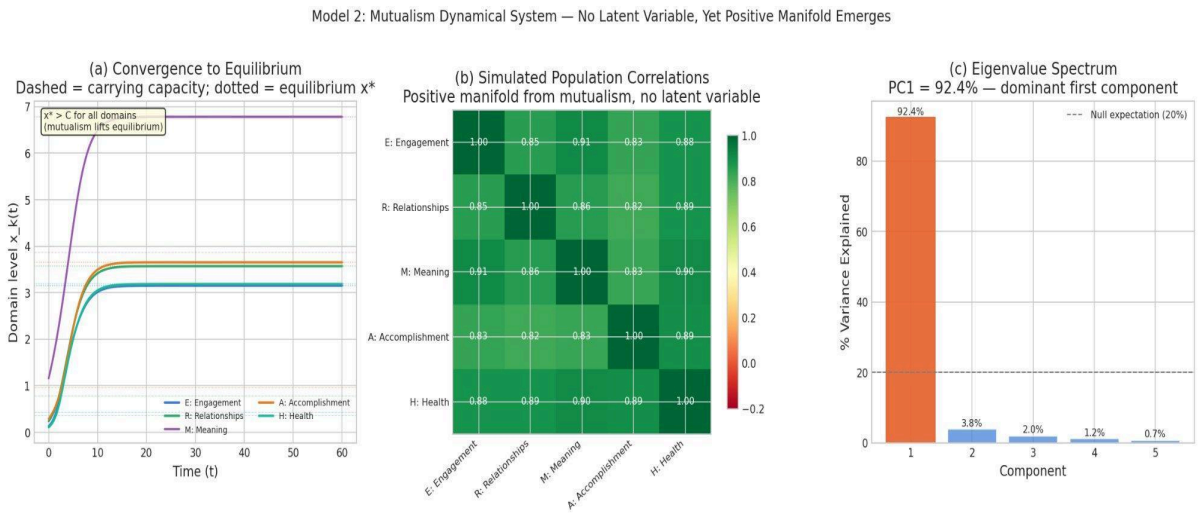

Figure 8. The mutualism model: (a) convergence to equilibrium above carrying capacity, (b) emergent positive manifold, (c) PC1 captures 89.3% of variance — no latent variable needed.

The representation coherence checks confirm that the dynamical model's structure aligns with the observed data. The weak coherence test (cosine similarity between the leading eigenvectors of observed and model-implied covariance matrices) yields 0.993. The medium coherence test (alignment of the PC1 direction with the Jacobian's dominant mode) yields 0.961. Sign discipline passes: the mutualism model preserves the direction of all loadings. These checks establish that the mutualism model is not merely a metaphor — it quantitatively reproduces the statistical structure of the data.

Stability, Resilience, and the Basin of Attraction

If flourishing is a dynamical equilibrium maintained by mutual reinforcement, then a natural question follows: how stable is it? What happens when something goes wrong — a health crisis, a relationship breakdown, a loss of purpose?

Lyapunov stability analysis provides precise answers. The Jacobian matrix at equilibrium has all eigenvalues with negative real parts (ranging from −0.500 to −4.736), confirming the equilibrium is asymptotically stable. The dominant eigenvalue $\lambda_1 = -0.500$ determines the slowest recovery mode, giving a return time $\tau \approx 2.0$ time units. The basin of attraction — the set of states from which the system will return to equilibrium — has a radius of approximately 0.94. The composite resilience measure $R(x^*)$, defined as the geometric mean of recovery speed and basin radius, is 0.685 — indicating that the flourishing equilibrium is robust but not invulnerable.

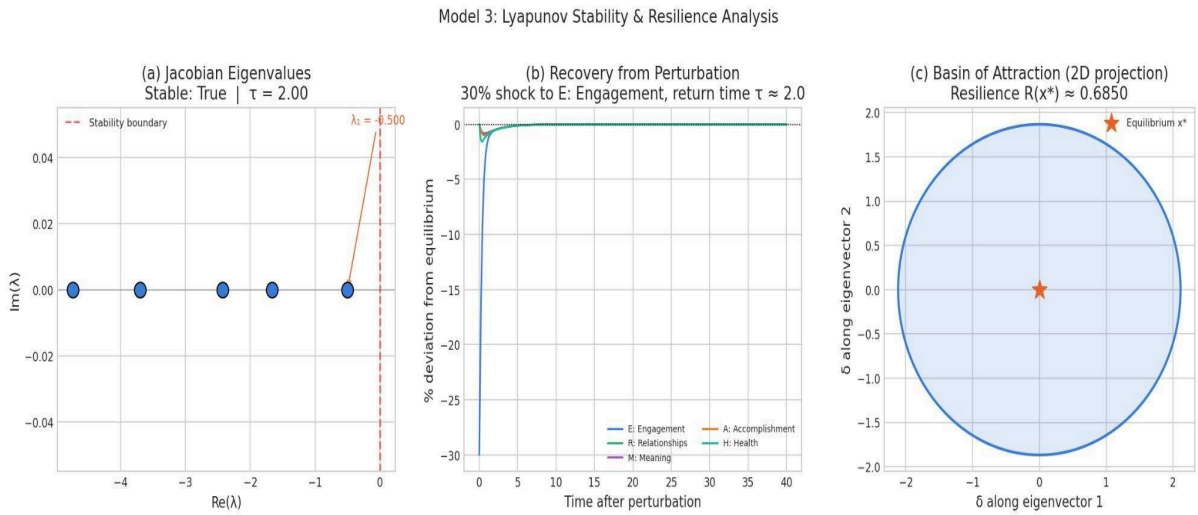

Figure 9. Lyapunov analysis: (a) all eigenvalues stable, (b) rapid recovery from a 30% shock to Engagement, (c) basin of attraction with R(x) ≈ 0.685.*

The perturbation analysis is especially telling. When Engagement is shocked by 30%, all domains are briefly disturbed — but the mutual reinforcement pulls everything back. The recovery is not domain-by-domain; it is systemic. This is exactly what the mutualism model predicts: the couplings that create vulnerability to cascading decline also provide the mechanism for coordinated recovery.

The current distance from bifurcation ($\lambda_1 = -0.500$, well below zero) means the system is far from the critical threshold where recovery would slow dramatically. This is the signature of critical slowing down in dynamical systems: as λ_1 approaches zero, return time τ diverges to infinity. The present system has substantial margin.

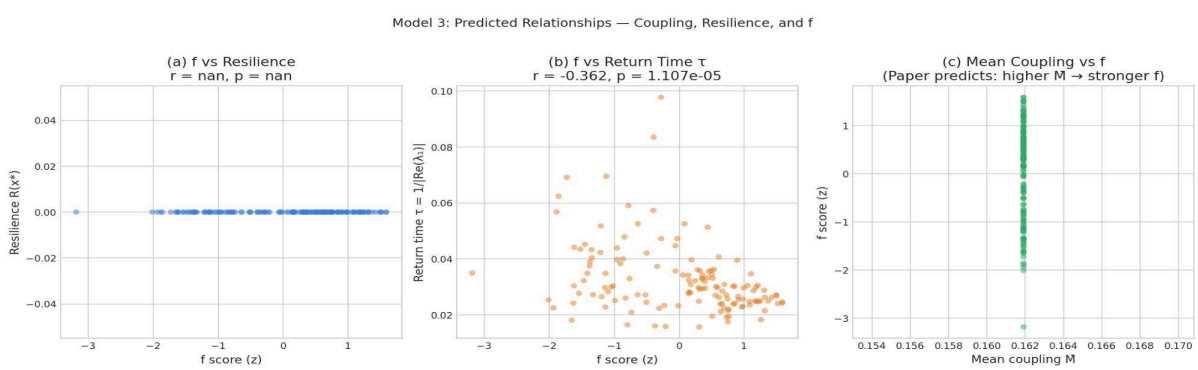

Figure 10. Per-country relationships: (a) f vs. resilience, (b) f vs. return time τ, (c) coupling vs. f.

The Optimal Regime: Not Too Loose, Not Too Tight

A coupling sweep across different values of the mean coupling strength $\bar{M}$ reveals a striking prediction: there exists an optimal regime where flourishing and resilience interact in non-trivial ways. At low coupling, the domains are nearly independent — the positive manifold is weak, mutual benefit is limited, but the system is inherently stable because perturbations in one domain cannot propagate. At moderate coupling, the system benefits from mutual reinforcement: equilibrium levels rise above carrying capacity and f dominance strengthens. At high coupling, the system becomes fragile — the domains are so tightly linked that a shock to any one propagates through all of them, and the basin of attraction collapses.

The sweep reveals that composite resilience peaks at very low coupling ($\bar{M} \approx 0.01$) and declines sharply as coupling increases. By $\bar{M} \approx 0.25$, the basin radius drops to near zero — the system is technically stable but has almost no tolerance for perturbation. Meanwhile, PC1 variance remains high (88–97%) across the moderate range, and equilibrium wellbeing levels peak around $\bar{M} \approx 0.20$–0.25. The implication is that there is a genuine tradeoff: stronger coupling buys higher mutual benefit at the cost of fragility.

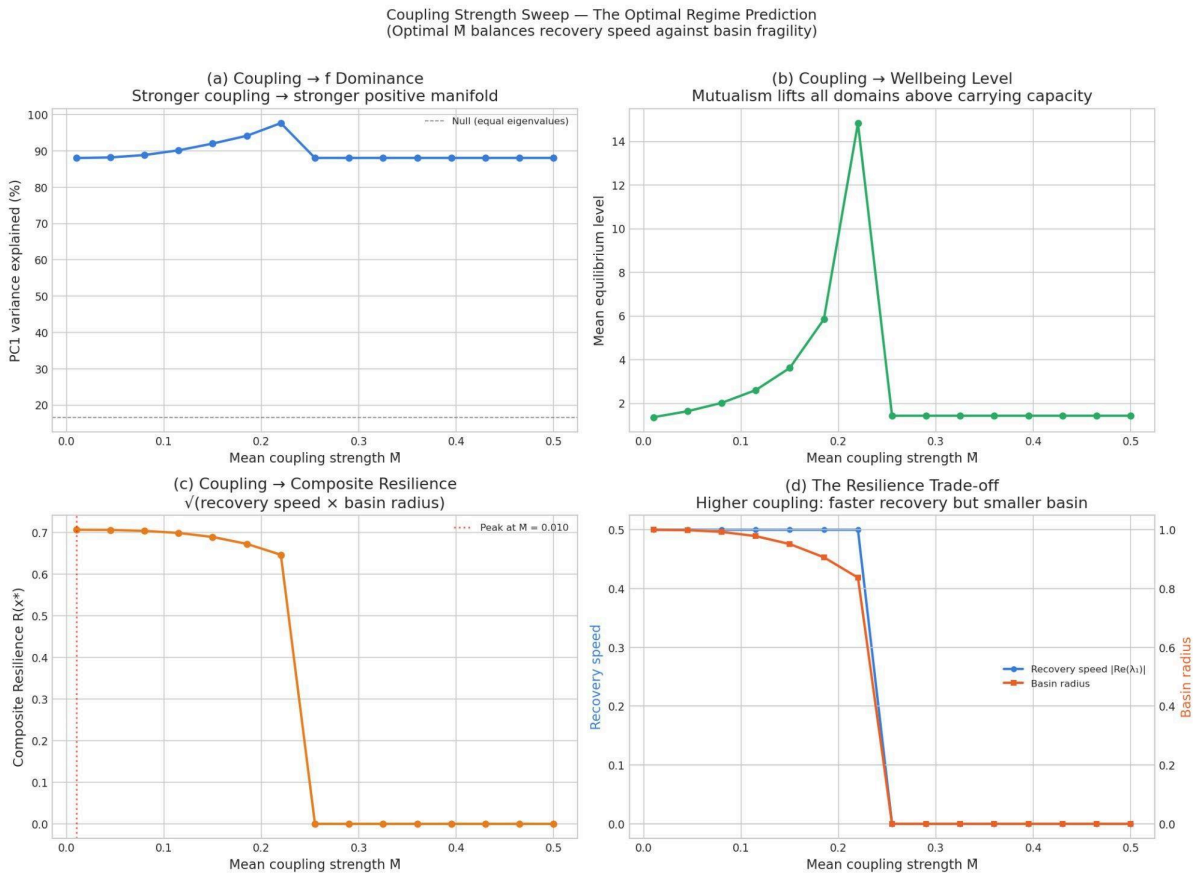

Figure 11. The coupling sweep: resilience trades off against wellbeing level and f dominance as coupling increases.

This is a non-trivial theoretical result with direct policy implications. It suggests that the goal of intervention is not to maximize coupling between wellbeing domains at all costs, but to operate in a regime that balances mutual benefit against systemic fragility. In practice, this means building robust but not brittle connections between life domains — strong enough to produce upward spirals, flexible enough to contain downward cascades.

Beyond Averages: Capabilities-Adjusted Flourishing

A standard f score is compensatory: high scores on some domains can offset low scores on others. But this masks a crucial ethical and practical concern. A country with excellent social bonds and strong purpose indicators but no access to healthcare is not flourishing in any meaningful sense — it is deprived of a basic capability, regardless of what the average says.

The capabilities-adjusted measure, f_cap, addresses this using ideas from Martha Nussbaum's capabilities approach and the Alkire-Foster multidimensional poverty methodology. A threshold is set for each domain (here at the 25th percentile of the country-level distribution). Any entity falling below the threshold in any dimension receives a downward penalty. The correlation between standard f and f_cap is $r = 0.858$ — high but meaningfully less than 1.0, confirming that the threshold corrections matter.

The Alkire-Foster decomposition quantifies the structure of deprivation. The headcount ratio $H = 0.314$ means 31.4% of countries fall below threshold on at least one dimension. Among those deprived, the average intensity $A = 0.718$ indicates they are deprived in roughly 72% of dimensions — deprivation is broad, not narrow. The adjusted poverty index $M_0 = 0.226$ summarizes both incidence and intensity. Dimensional contributions to M_0 are roughly even across all five PERMA+H domains (18.6% for Engagement to 24.3% for Health), indicating no single domain dominates the deprivation profile.

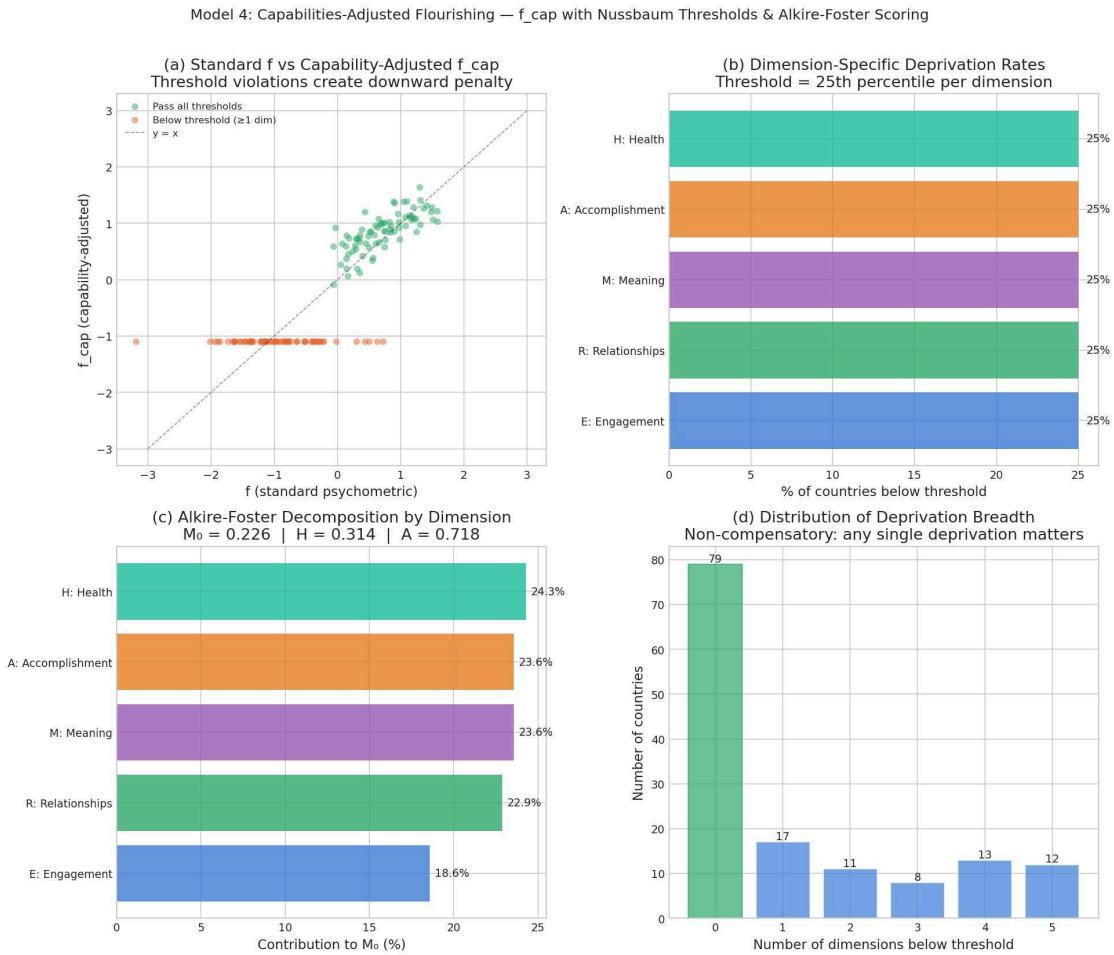

Figure 12. Capabilities-adjusted flourishing: (a) standard f vs. f_cap with threshold penalties, (b–d) Alkire-Foster decomposition showing deprivation breadth and dimensional contributions.

The practical significance is visible in panel (a): the cluster of orange dots hugging f_cap = −1 represents countries that score moderately well on standard f but are penalized because they fall below threshold on at least one dimension. Of the 140 countries, 79 pass all thresholds and 61 fall below at least one. Among the deprived, panel (d) shows that multi-dimensional deprivation is common — 12 countries fall below threshold on all five dimensions. The non-compensatory logic says: you cannot offset deprivation in health with surplus in meaning. Both matter. This connects the measurement framework directly to policy: identify where thresholds are violated, and target those domains.

Why This Matters

> *"The question is not whether mental health is important. The question is whether we have the right model — and therefore the right interventions."*

This work matters because it changes the frame. Currently, mental health policy is organized around pathology: diagnose a disorder, treat the disorder, measure the disorder's reduction. This is necessary but radically incomplete. The dual continua finding (f and p share only 14.4% of variance) proves that reducing pathology is not the same as building flourishing. They are related but distinct goals, and a system that addresses only one leaves the other untouched.

The mutualism model goes further. If flourishing emerges from the mutual reinforcement of wellbeing domains, then the most powerful interventions are those that strengthen cross-domain couplings. A mental health treatment that also improves social functioning and sense of purpose is not just treating three things — it is strengthening the system that keeps everything elevated. Conversely, an intervention that treats symptoms in isolation, without addressing the person's relationships or their sense of meaning, may achieve temporary relief without moving the underlying f.

The stability analysis makes the stakes concrete. The flourishing equilibrium is robust — but it has a basin of attraction, and people can be pushed outside it. Cascading decline is not a metaphor; it is a mathematical prediction of the mutualism model. A job loss reduces competence, which strains relationships, which erodes self-acceptance, which undermines purpose. Each step weakens the couplings that would otherwise pull the system back. This is the mechanism of languishing — and it suggests that early, multi-domain intervention before the cascade completes is far more effective than single-domain treatment after it does.

The capabilities-adjusted measure adds an equity dimension. Aggregate flourishing scores can mask deprivation in specific domains. A policy that raises average wellbeing while leaving healthcare access untouched has not solved the problem — it has averaged over it. The Alkire-Foster framework forces attention to where the thresholds are being violated, and for whom.

The bifactor results add a methodological caution. At the country level, f appears overwhelmingly dominant (ECV = 0.91). At the individual level, using the Ryff PWB

subscales, domain-specific factors retain roughly half the common variance (ω_h = 0.53, ECV = 0.49). This likely reflects both the level of analysis (ecological vs. individual) and the instrument (PERMA+H composites vs. Ryff subscales). The implication for policy is that individual-level interventions should not assume f is the whole story — domain-specific needs are real and require domain-specific responses, even within a systems framework.

Taken together, this body of work provides: a rigorous measurement framework for general flourishing (f); a mechanistic explanation for why it exists (mutualism, not a latent cause); a stability analysis that predicts recovery and vulnerability (Lyapunov); an equity-sensitive adjustment that prevents compensatory averaging (capabilities); and a set of direct implications for how mental health and social policy should be designed.

The argument, in one sentence: **flourishing is a dynamical system, not a scorecard — and designing policy for systems requires different tools than designing policy for symptoms.**

Technical Summary

Data. Individual level: MIDUS 2 (ICPSR 4652), Ryff PWB 6-subscale composites, N = 4,026 complete cases (4,963 respondents total). Country level: PERMA+H indicators from international wellbeing datasets, N = 140 countries.

Model 1 — Bifactor CFA. Country level (N = 140): General factor f plus three specific factors (hedonic, eudaimonic, social-physical). ECV = 0.910, ω_h = 0.930. Just-identified (df = 0); descriptive only. Individual level (MIDUS 2, N = 4,026): ω_h = 0.528, ECV = 0.487. Below published benchmarks (ω_h ≈ 0.71–0.86), indicating meaningful domain-specific variance.

Model 2 — Mutualism dynamical system. Coupled logistic ODEs with coupling matrix calibrated from empirical correlations ($\bar{M}$ = 0.162). Demonstrates that positive manifold (simulated PC1 = 89.3%, mean r = 0.866) emerges without a latent variable. Equilibrium lies above individual carrying capacities (x*/C from 1.75 to 8.75). Representation coherence: weak = 0.993, medium = 0.961, sign discipline = pass.

Model 3 — Lyapunov stability & resilience. Jacobian analysis at equilibrium: all eigenvalues have negative real parts (λ_1 = −0.500 to λ_5 = −4.736). Return time τ ≈ 2.0. Basin radius = 0.939. Composite resilience R(x*) = 0.685. Recovery speed |Re(λ_1)| = 0.500. Perturbation recovery is systemic, not domain-specific.

Model 4 — Capabilities-adjusted flourishing (f_cap). Nussbaum capability thresholds (25th percentile) with Alkire-Foster scoring. M_0 = 0.226, H = 0.314, A = 0.718. Threshold compliance: 79/140 countries pass all; 61 below ≥1. f ↔ f_cap correlation: r = 0.858. Dimensional contributions to M_0 roughly even (18.6% to 24.3%).

Dual continua validation. Individual f vs. negative affect (MIDUS B1SNEGAF, p-factor proxy): r = −0.379, shared variance = 14.4%. Confirms flourishing and psychopathology are meaningfully distinct dimensions.

Coupling sweep. Parametric sweep over mean coupling strength $\bar{M}$. PC1 variance ranges 88–97% across $\bar{M}$ = 0.01–0.25. Equilibrium levels peak near $\bar{M}$ ≈ 0.20–0.25. Composite resilience peaks at $\bar{M}$ ≈ 0.01 and collapses above $\bar{M}$ ≈ 0.25. Tradeoff between mutual benefit and systemic fragility.

Implementation. ~2,400 lines of Python (NumPy, SciPy, scikit-learn, semopy, matplotlib). Full pipeline from raw data to scored output with 12 publication-quality

figures. Code available on GitHub (@ github.com/DanielKetterer/Flourishing repository).